

Ian Burress

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EDUCATION

Texas A&M University | Master of Science in Mechanical Engineering

May 2024

- M.S. Thesis Title: *"The Design of a Mobile Electron Beam Treatment Station For Contaminated Soils"*

Michigan State University | Bachelor of Science in Mechanical Engineering

May 2022

- *NCAA Wrestler*, Michigan State; *Senior Business Director*, Solar Car Racing Team; *Research Assistant*, College of Engineering

EXPERIENCE

Test & Development Engineer - Rotational

July 2025 - Present

Caterpillar | Clayton, NC

- Performed tests operating software including Dewesoft, CANape, and CADet with equipment including strain rosettes, displacement yoyos, thermocouples, pressure transducers, instrumented bolts, and phototachs for analysis transmitted CAN bus
- Reinforced learning with on site Geometric Dimensioning and Tolerancing (GD&T) training, mastering core principles of engineering tolerances and precision manufacturing
- Oversaw, executed or contributed 3 test plans, data analysis, and failure investigations using industry-standard methods

Circuit of The Americas Fire Marshal

March 2024 - Present

Circuit of The Americas | Austin, TX

- Served as Certified SCCA Pit Lane Intervention Marshal at major motorsports events (World Endurance Championships, NASCAR Cup Series, Mustang Challenge, Lamborghini Super Trofeo N.A.)
- Ensured safety for drivers, crew, and spectators by monitoring pit lane, manning fire extinguishers during refueling, and enforcing relevant safety protocols
- Required for fire suppression during fire during 2024 Ford Mustang Challenge, tire caught fire due to wheel lockout

Solar Car Dynamics and Body & Sizing Judge

July 2025 - Present

Innovators Educational Foundation | Rolla, MO

- Race official for North American racing series hosting 50+ collegiate teams in a closed track and cross country endurance race
- Directly responsible for making sure all regulations regarding interior sizing and vehicle exterior meet track and road standards and qualifications
- Core group of race officials running pre race dynamic assessment qualification events including slalom, figure 8, wet brake, acceleration test, and u-turn test. Used these tests to assess road worthiness of student design vehicles
- Review and assist teams to ensure race ready compliance prior to event via design reviews and open lines of communication

G3500 Platform Engineer - Rotational

March 2025 - July 2025

Caterpillar | Lafayette, IN

- Rebuilt field failure engine from block-up; organized 360+ components across 17 kits and resolved 28 part discrepancies
- Investigated turbocharger failures across large engine platforms, assisted in teardown of 3 customer-returned units and documented failure modes for ongoing root cause analysis
- Documented roading and bump tests of customer designed chassis; collected engine performance data at multiple operating points

Product Support Engineer - Rotational

December 2024 - March 2025

Caterpillar | Lafayette, IN

- Conducted warranty investigations and failure analysis in Large Power Systems Division (LPSD) for marine, locomotive, oil and gas, and electric power industries
- Interfaced dealers and customers in root cause analysis of in-field failures, tracing issues back to failed components
- Facilitated various New Product Introduction (NPI) projects from inception from warranty level through handoff to platform engineering team

Application and Installation Engineer - Rotational

August 2024 - December 2024

Caterpillar | Houston, TX

- Executed critical validation fieldwork, including 6 in field audits with multiple days of prior equipment troubleshooting
- Served as corporate liaison between global dealers and customers, facilitating effective communication, resolving technical inquiries, and ensuring alignment with Caterpillar's product support strategies
- Conducted a comprehensive audit for company and dealer wide conference, categorizing prior-year data to strategic touchpoints

Manufacturing Technology & Solutions Intern

May 2023 - August 2023

Caterpillar | Chillicothe, IL

- Earned Six Sigma Green Belt leading DMAIC process to manage safety related improvement projects
- Developed an installation plan, for magnetic sensing of cabling for crane technologies, saving \$250,000 if implemented

Graduate Research Assistant

August 2022 - May 2024

Texas A&M University | College Station, TX

- Worked under Dr. David Staack and Department of Energy to complete a thesis design of a mobile radiation treatment facility for contaminated soil treatment
- Hosted weekly design reviews with research advisor to review project and strategize potential avenues of improvement
- Deployed Solidworks to gather data such as feasibility, facility weight, and FEA simulations

Integrated Components & Solutions Intern

May 2022 - August 2022

Caterpillar | Peoria, IL

- Facilitated adoption of assisted reality into manufacturing environment. Activities included proof of concepts of Teamviewer product for work instructions, quality gates, gathered feedback to understand time usage and data inputs
- Oversaw production, contributed to fabrication troubleshooting with operators, provided feedback for enhanced end quality

Research Assistant

November 2020 - May 2022

Michigan State University | East Lansing, MI

- Worked under Dr. Elisa Toulson Measured and recorded laminar flame characteristics of sequential two stage combustion
- Processed and evaluated images captured from experimental apparatus through a high speed camera operating a Schlieren Light source
- Leveraged Matlab to gather data such as radius, area, and circularity to find laminar flame expansion speed

Manufacturing Quality Intern

May 2021 - August 2021

Stellantis | Dundee, MI

- Initiated 2 CPIs, managed plant-wide audits, and evaluated 15 stations for defects deploying World Class Manufacturing strategies and databases such as EBOM and MQAS
- Produced 5+ Kaizens enhancing quality and saving over \$50,000 annually in production and warranty costs

Interior Design Intern

June 2020 - August 2020

Fiat Chrysler Automobiles | Auburn Hills, MI

- Hosted Q&A referencing knowledge of Autodesk, Photoshop, and various CAD softwares to build visualizations, collaborated cross functionally with 20+ other employees to ensure feasibility of an infotainment design concept
- Awarded most innovative concept, FCA to research idea for implementation and patent for future vehicles

SKILLS

CAD: Siemens NX, Autodesk, SolidWorks, Creo

Analysis: Thermal Desktop, Ansys , LS-DYNA, STAR-CCM+

Languages: Python, MATLAB, Microsoft Excel VBA